

Problem Set for 3rd March 2025

Exercise: Determine linear maps $\alpha, \beta : V \rightarrow V$ such that

$$\alpha \circ \beta = \text{id}_V, \quad \text{but} \quad \beta \circ \alpha \neq \text{id}_V.$$

答: 令 $V = \prod_{k \in \mathbb{N}} \mathbb{F}$, 即 $\mathbb{F}$ 中取值的数列空间. 定义

1. (左移) $L : (x_0, x_1, x_2, \dots) \mapsto (x_1, x_2, x_3, \dots)$;
2. (右移) $R : (x_0, x_1, x_2, \dots) \mapsto (0, x_0, x_1, \dots)$.

此时, $L \circ R$ 是恒等映射, 但 $R \circ L$ 不是恒等映射.

Problem: Let $\mathbb{F}$ denote the ground field, and let S be any finite set.

1. Demonstrate that $\text{Hom}_{\text{Sets}}(S, \mathbb{F})$ forms a vector space.

答: 若 f 与 g 是 $(S \rightarrow \mathbb{F})$ -类型的映射, 则对任意 $\lambda \in \mathbb{F}$,

$$(\lambda f + g) : S \rightarrow \mathbb{F}, \quad s \mapsto \lambda(f(s)) + g(s)$$

也是 $(S \rightarrow \mathbb{F})$ -类型的映射. 这说明 $\text{Hom}_{\text{Sets}}(S, \mathbb{F})$ 对线性组合封闭, 从而是线性空间.

2. Construct a linear bijection (hereinafter referred to as a *linear isomorphism*)

$$\text{Hom}_{\text{Sets}}(S, \mathbb{F}) \rightarrow \mathbb{F}^{|S|}.$$

答: 若 S 空, 则左右两侧都是一元集. 也就是零线性空间. 假定 S 非空, 记 $S = \{s_i\}_{1 \leq i \leq n}$, 取 ϕ 是如下映射

$$\phi : \text{Hom}_{\text{Sets}}(S, \mathbb{F}) \rightarrow \mathbb{F}^{|S|}, \quad f \mapsto (f(s_1), \dots, f(s_n)).$$

ϕ 将 f 映至其定义式 $\{s \mapsto f(s)\}_{s \in S}$, 从而 ϕ 是单的; 任意 $(v_s)_{s \in S} \in \mathbb{F}^{|S|}$ 都能定义函数 $\{s \mapsto v_s\}_{s \in S}$, 从而 ϕ 是满的.

3. Demonstrate that following function constitutes an injection of sets:

$$\varphi : S \rightarrow \text{Hom}_{\mathbb{F}}(\text{Hom}_{\text{Sets}}(S, \mathbb{F}), \mathbb{F})$$

$$s \mapsto \begin{bmatrix} \text{Hom}_{\text{Sets}}(S, \mathbb{F}) & \rightarrow & \mathbb{F} \\ f & \mapsto & f(s) \end{bmatrix}$$

答: 用自然语言表述, φ 将 $s \in S$ 映至线性函数在 s 处的赋值. 为证明 φ 单, 只需证明 $\varphi(x) = \varphi(y)$ 蕴含 $x = y$. 若不然, 假定 $x \neq y$, 则任取满足 $f(x) \neq f(y)$ 的函数 $f : S \rightarrow \mathbb{F}$. 此时,

$$(\varphi(x))(f) = f(x) \neq f(y) = (\varphi(y))(f).$$

这与 $\varphi(x) = \varphi(y)$ 矛盾.

4. Demonstrate that the image $\varphi(S)$ forms a basis for $\text{Hom}_{\mathbb{F}}(\text{Hom}_{\text{Sets}}(S, \mathbb{F}), \mathbb{F})$.

答: 取函数 $\{f_i\}_{1 \leq i \leq n} \subset \text{Hom}_{\text{Sets}}(S, \mathbb{F})$, 满足

$$f_i : s_j \mapsto \delta_{i,j} \cdot 1 \quad (1 \text{ 是域 } \mathbb{F} \text{ 的单位元}).$$

以上定义的 $\{f_i\}_{1 \leq i \leq n}$ 的线性无关, 因为对任意线性组合式 $\sum c_i f_i = 0$, 总有

$$\varphi(s_j) : \sum c_i f_i \mapsto c_j = 0.$$

比较维数, $\{f_i\}_{1 \leq i \leq n}$ 是 $\text{Hom}_{\text{Sets}}(S, \mathbb{F})$ 的一组基, 其对偶基是 $\{\varphi(s_i)\}_{1 \leq i \leq n}$.

5. This is how we define

$$\mathbb{F}s_1 \oplus \mathbb{F}s_2 \oplus \cdots \oplus \mathbb{F}s_n \quad S = \{s_1, \dots, s_n\}.$$